MIL-T-38306(USAF)
AMENDMENT 4
22 SEPTEMBER 1983
SUPERSEDING
AMENDMENT 3
03 August 1983

MILITARY SPECIFICATION

TRANSMITTER, RATE GYROSCOPE T-970/AJB-7

This amendment forms a part of Military Specification, MIL-T-38306(USAF), dated 31 July 1964 and is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

Page 1

2.1: Delete paragraph entirely and all specifications/standards. Add new paragraph as follows:

"2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-636	-	Box, Shipping, Fiberboard

Military

MIL-P-116	-	Preservation, Methods of
DOD-D-1000	-	Drawings, Engineering and Associated Lists
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-I-6181	-	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-I-27619	-	Indicator, Attitude Director
MIL-C-38295	-	Computer Set, Attitude-reference Bombing AN/AJB-7
MIL-C-83488	-	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

Military

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

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MIL-STD-143	-	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-470	-	Maintainability Program Requirements (For Systems and Equipment)
MIL-STD-471	-	Maintainability Demonstration
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-781	-	Reliability Test Exponential Distribution
MIL-STD-785	-	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	-	Environmental Test Methods
MIL-STD-831	-	Test Reports, Preparation of
MIL-STD-889	-	Dissimilar Metals

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Copies of specifications, standards, handbooks, drawings, and publications, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)"

Add the following boxed paragraph to bottom of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressd to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

3.3.5: Delete "MS33586" and substitute "MIL-STD-889".

PAGE 3

3.4.1: Delete last sentence of paragraph.

3.4.2: Delete paragraph and substitute the following:

"3.4.2 Reliability. The transmitter shall be so designed that when tested as specified in 4.6.26, the upper test mean-time-between-failure (θ_o) will be 5,000 hours. The lower test mean-time-between-failure (θ_1) shall be 1667. The test program shall be based upon a discrimination ratio of 3.0 and a confidence level of 90 percent. MIL-STD-785 shall be used as a guide for proper implementation of the required reliability into the design and production of the equipment."

3.4.2.1: Delete paragraph and substitute the following:

"3.4.2.1 Longevity. The transmitter shall have an operating life span of not less than 2,000 hours, before the failure rate increases or wearout failures occur."

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PAGE 4

3.5.2, line 2: Delete "90" and substitute "120".

PAGE 5

3.5.6: Change "5 cps" to "5 Hz".

3.5.9, line 2: Delete "1.75 ± 0.25" and substitute "0.24 ± 0.10".

PAGE 6

3.6.1, line 2: Change "400-cps" to "400 Hz" twice. Line 5, change "cps" to "Hz".

3.6.3.1, lines 5 and 9: Delete "400 cps" and substitute "400 Hz".

3.6.5, line 1: Change "damped" to read "damaged".

3.7, line: Delete "precession" and substitute "input".

PAGE 7

3.7.2: Delete second sentence and substitute: "The filling medium shall be either 100 percent helium, 95 percent helium and 5 percent carbon dioxide, or a mixture of 88 to 92 percent nitrogen and the remainder helium.

3.8, line 4: Delete "MIL-D-70327" and substitute "DOD-D-1000".

3.10: Delete entirely.

PAGE 8

3.14.1.5, line 1: Change "Whem" to read "When".

Add new paragraph 3.14.1.6 as follows:

"3.14.1.6 Parts enclosed in a hermetically sealed container need not be anodized or plated."

3.14.2: Delete and substitute the following:

"3.14.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, in accordance with MIL-C-83488, Type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

3.16.1, line 2: Change "consist" to "consistent".

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PAGE 11

4.4.1.1: Delete paragraph and substitute the following:

"4.4.1.1 Reliability demonstration. Unless otherwise specified, 11 transmitters representative of the production equipment produced on the contract or order shall be tested as specified in 4.6.26 for a total test time of 17,250 hours, unless rejection is determined in accordance with figure 2 prior to that time. Acceptance should occur at 17,250 hours if no more than 6 failures have occurred."

PAGE 12

*Figure 2: Delete and substitute new figure 2.

PAGE 15

4.5.3: Delete and substitute the following:

"4.5.3 Sampling plan C. Unless otherwise specified, 9 transmitters shall be selected at random from each 50 or less produced on the contract or order and tested as specified in 4.6.26 in accordance with Test Plan VC of MIL-STD-781C."

4.5.3.1: Delete and substitute the following:

"4.5.3.1 Acceptance/rejection. Certified time records shall be maintained of the test time accumulated on the equipment after starting the environmental cycling specified in 4.6.26. The accept/reject criteria of Test Plan VC of MIL-STD-781C will be used."

4.5.3.2: Delete "MIL-R-26667" and substitute "MIL-STD-781".

PAGE 16

Figure 3: Delete figure 3 entirely.

PAGE 17

4.6.5, line 4: Delete "1.47" and substitute "0.37", and delete "1.75 ± 0.25" and substitute "0.24 ± 0.10".

4.6.6, second sentence: Delete and substitute "The output reading shall not change more than 10 ua from null."

4.6.8, line 3: Between "terminals" and "for", insert "and between isolated terminals and the case".

4.6.10, line 2: Delete "more" and substitute "less".

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PAGE 18

4.6.16.1: Delete entirely.

PAGE 19

4.6.18: Change "method 501" to "method 501.1, procedure II".

4.6.19: Change "method 504, class 2" to "method 504.1, equipment category 6," change "in step 5" to "in step 6" and change "in step 7" to "in step 8".

4.6.20: Change "method 503" to "method 503.1".

4.6.21: Change "method 516, procedures I and IV" to "method 516.2, procedures I and IV".

4.6.22: Change "method 514, curve B" to "method 514.2, category b. 2".

PAGE 20

4.6.24.1: Change "method 507" to "method 507.1".

4.6.24.2: Change "method 508, procedure I" to "method 508.1".

4.6.24.3: Change "method 509, procedure I" to "method 509.1".

4.6.24.4: Change "method 510, procedure I" to "method 510.1".

4.6.26.1: Delete and substitute the following:

"4.6.26.1 Longevity. At the completion of the test specified in 4.6.26, three of the transmitters shall continue to be tested for a total of 2,000 hours each. The evaluation shall be limited to the analysis of all failures, identification of design and pattern failures and the submission by the contractor of recommended design changes to the procuring activity."

4.6.26.4: In second sentence, delete "in accordance with MIL-R-26667".

4.6.27: Delete and substitute:

"4.6.27 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471A, Test Method 9."

Add new paragraph as follows:

"4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

Delete Section 5 entirely and substitute the following:

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"5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2,d).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitters shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Transmitters shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with method IC-1 of MIL-P-116. Each transmitter shall be overboxed in PPP-B-636 box. Carton shall be large enough to allow for sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak valves in g's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Transmitters shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage, during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C (as specified in 6.2,f).

5.2.1 Level A. Transmitters packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Style A or B, class overseas, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniform shape, size, and of minimum tare and cube, consistent with the protection required.

5.2.2 Level B. Transmitters packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129."

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Air Force-99

Preparing activity:
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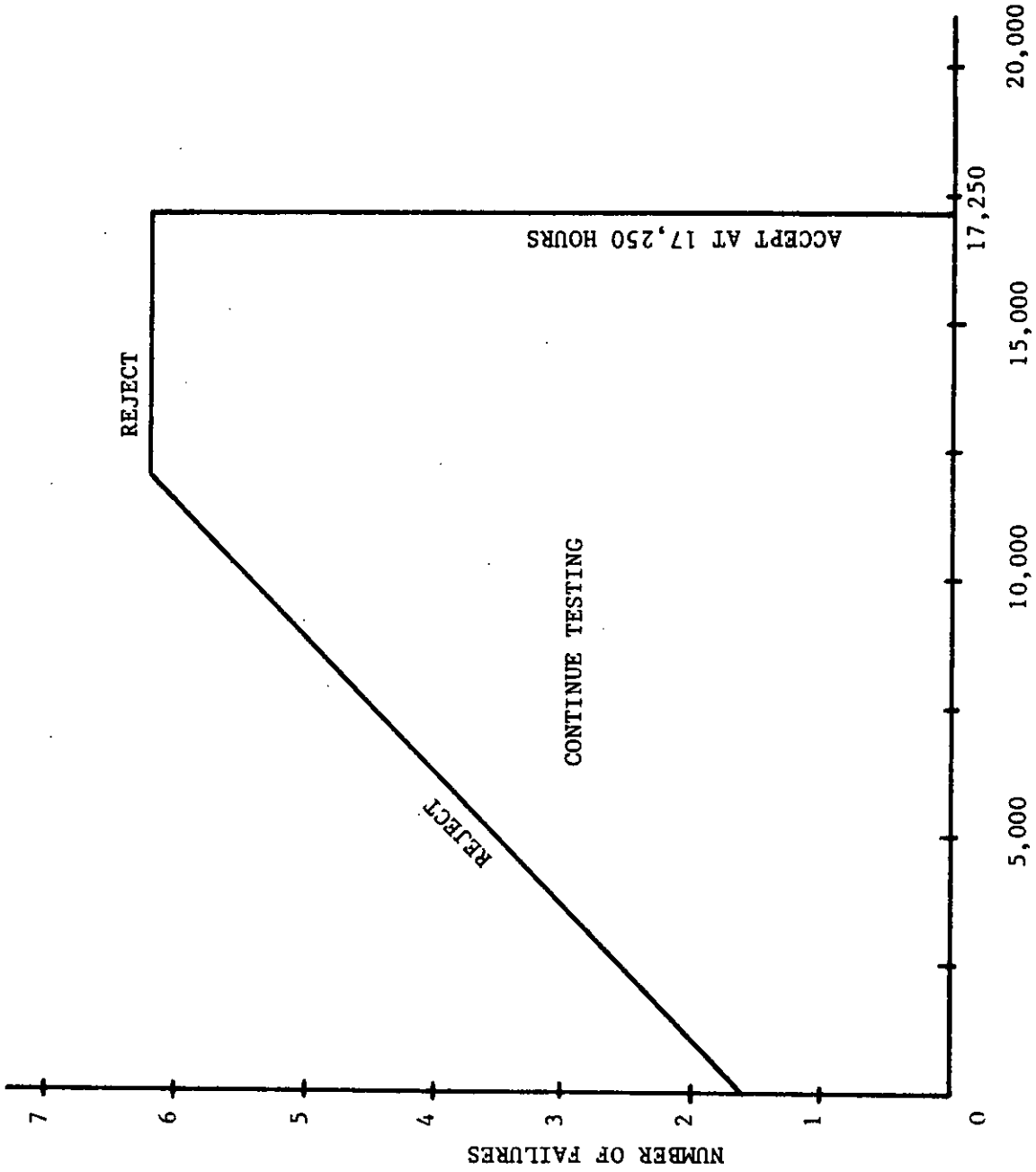


FIGURE 2. Continue test-reject criteria

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